A Self-Interpreter for SKI Written in SKI: Extensional Semantics for New Symbols

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Abstract

A symbol added to an object language has no meaning until an interpreter gives it one. We make that authorship visible at the smallest scale we can find: a self-interpreter for SKI combinatory logic written as a closed term of the basis $\{S, K, I\}$, 618 atoms, which runs on an encoding of itself in 504,930 contractions. Extending the object language with a fifth symbol forces the interpreter's result type to grow, and whether the new outcome is exposed as a value or absorbed into divergence is decided by a single arm of the interpreter, a few atoms at one named site, with the two variants provably agreeing on every term that never reaches it. The same site, made a parameter, becomes a nonlocal lookup in the style of Nock's scry: the resolver is passed as an argument, a lookup may block, and a namespace of answers accumulates across re-runs, keyed by a 240-atom structural equality on encoded paths. Two structural results frame the construction. A census over the $\{B, C, K, W\}$ basis shows that representation needs weakening, that iteration needs contraction, and that contraction alone already costs normalization. And the one step no SKI term can perform is quotation of a live term, by a theorem of Jay and Given-Wilson; above that boundary everything, including metacircular evaluation, is closed.

1 Introduction

Metacircular evaluation permits the separation of the evaluation mechanism from the object language, allowing for controlled experimentation with different evaluation strategies and the observation of intermediate states without affecting the host environment. An interpreter can be employed to handle states such as partial evaluation without interfering with the host environment, particularly in cases in which the host environment itself may not be capable of representations such as partially evaluated terms, intermediate representations of the object language, crashes, and other exceptional states.

We construct a symbolic step-evaluator for the SKI combinator calculus as a closed term of the basis $\{S, K, I\}$. This evaluator performs symbolic evaluation by repeatedly applying the reduction rules of the SKI calculus until a normal form is reached. Because the interpreter is capable of symbolic representation beyond simply the basis $\{S, K, I\}$, we can add in functionality such as handling additional combinators, supporting custom reduction strategies, and managing exceptional states like crashes, undefined terms, or as-yet-unknown referentially transparent lookups.

The current line of research is motivated by the question of how amenable the SKI combinator calculus is to metacircular evaluation, and what design patterns and implementation techniques could facilitate such evaluation. We do not anticipate such a line of inquiry to

be competitively efficient versus other combinator systems or higher-level representations, but rather as a tool for further exploration of the metacircular evaluation process and its possibilities for simple programs. This paper makes three contributions:

1. A self-interpreter for $\{S, K, I\}$ in $\{S, K, I\}$. This evaluator is a step-evaluator over encoded terms rather than a decoder, together with the structural census that says what each rule of the calculus is bought for (§3, §4).
2. The observation that extending the object language forces the interpreter's outcome type to grow, and that the meaning of the new outcome is authored at one site, with a lemma that nothing else changes (§5).
3. Site as parameter. This is a value lookup whose resolver is an argument, which can block, and whose namespace of answers grows across re-runs over structurally compared paths (§6).

Section 2 places the work in the metacircular tradition; §7 reads the construction against virtualization in the Nock combinator calculus. The artifact and its reproduction are described in Appendix A.

2 Metacircular evaluation and self-interpretation

While many languages nominally support it, metacircular evaluation as a first-class strategy has only occasionally been applied outside of the Lisp family, where it has been used to implement features such as macro systems, partial evaluation, and sandboxed execution environments. Essentially, each nested interpreter acts as a virtual machine for the next level of interpretation. Lisp's homoiconic nature is particularly apt for such virtualization, since the interpreter can directly treat its interpretand as data, enabling Lisp's powerful reflective and metaprogramming capabilities. (In this connexion, the collapsing towers of Amin and Rompf [2018] will be of special interest.)

The lineage runs from McCarthy [1960], whose universal function `eval` was written in the language it evaluates, and in particular from the Lisp 1.5 manual [McCarthy et al., 1962], which specifies the language by that function. Reynolds [1972] gave the form its critical treatment: a definitional interpreter written in the style of its own object language is metacircular; the interpreted language inherits the defining language's evaluation order unless that order is made explicit. (The present work meets that observation directly in §5.1, where the object-level laziness of K is inherited rather than implemented.) Guy L. Steele and Sussman [1978] and Abelson et al. [1996] made the metacircular evaluator a standard instrument for experimenting with semantics, the very use to which we put it.

The Nock combinator calculus (est. 2002–2006) explicitly adopted self-application as a fundamental mechanism for metacircular evaluation from its earliest inception [Yarvin, 2018]. This mechanism was originally intended as a "hyper-Turing" operator [Davis, 2025], in which a query for a referentially transparent value could answer directly, with a permanent refusal, or with a deferral.

The tower as such entered with Smith [1984], whose 3-Lisp runs each level of a reflective tower on an interpreter at the level above; Wand and Friedman [1988] gave the tower a non-reflective description, and Asai et al. [1996] defined the language Black (in whose style Amin and Rompf's Purple is cast). Amin and Rompf [2018] treated the tower itself as the object of study. Their multi-level λ-calculus provides staging constructs and stage polymorphism, so that a single evaluator acts as an interpreter or as a compiler depending on runtime parameters, and

their metacircular evaluator Pink collapses arbitrarily many levels of self-interpretation into one compiled program; correctness of self-interpretation and multi-level Jones optimality are stated without formal proof but backed by experiment (their Propositions 4.2 and 4.6). Semantic modifications introduced at any level of the tower, such as instrumentation, CPS conversion, or reflection in the style of Black, survive the collapse. Their objective is performance, the removal of interpretive overhead, and their semantics is value-producing throughout: there is no fuel, no failure, and no outcome that has to be represented rather than computed. The present work complements their approach. We retain the levels with payment (§7), and we ask what an interpreter does in the case where its object language gains a construct whose meaning the interpreter must author (§5); this question does not arise in a setting without a third outcome. The two share the shape of the mechanism: an interpreter as an ordinary program in the calculus it interprets, with one place at which its behaviour is substituted. §6 builds upon this foundation.

In the λ-calculus proper, Barendregt [1991] stated the requirement on a self-interpreter, $\mathbf{E}\ulcorner M\urcorner = M$, and notes that the coding $\ulcorner\cdot\urcorner$ is a metalevel operation rather than a term; that boundary is located by a theorem of Jay and Given-Wilson [2011] for SKI (§3.3). Berarducci and Böhm [1993] show that the choice of encoding is load-bearing, obtaining a self-interpreter that itself has a normal form, and Brown and Palsberg [2016] carry self-interpretation into System F_ω, in which typing rather than quotation forms the barrier. Futamura's first projection [Futamura, 1971] handles the specialization of an interpreter to a fixed program; Jones et al. [1993] provide an standard account of this work. Jones [1996] identified the optimality condition which must be met by a specializer to remove interpretive overhead entirely.

Self-interpretation for a Turing-complete system is as old as universality: Kleene [1936] gave the normal form theorem from which a universal function follows, and Turing [1936] the universal Mogensen [1992] built a compact self-interpreter and self-reducer in the untyped λ-calculus; Jay and Palsberg [2011] built a typed one, in a combinatory calculus with a factorisation operator though not in $\{S, K\}$. (In particular, their distinction between self-*recognisers*, which recover a term from its representation, and self-*enactors*, which execute it, is necessary here.) The closest prior construction for SK is Tromp [2006], whose Theorem 3 gives a λ-term F of size 124 with $F\,C\,(\tilde{M} : N) = C\,M\,N$ for a bitstring encoding $\tilde{M}$ of a combinator M, and whose Theorem 4 translates it into a 263-bit self-interpreter for combinatory logic; with $C = I$ it is a self-enactor. Architecturally F is a decoder: it reconstructs live syntax from bits and hands it to a continuation, after which reduction proceeds under the ambient calculus's own rules. It carries no object-level S or K rule in its body, and so cannot implement a step relation that differs from the host's, which is what we carry out in §5. The same paper contains a symbolic normal-form reducer over Scott-encoded de Bruijn terms for the λ-calculus, used only as a size constant.

Our novelty is narrow; a symbolic step-evaluator for SKI in the λ-calculus is a textbook exercise, and bracket abstraction carries it into $\{S, K, I\}$ mechanically. What we add is a concrete fueled expression of 618 atoms with its step rules as data manipulation rather than re-spliced host syntax, applied to an encoding of itself and run to completion. On top of this we add a census (§4), the one-site authorship and faithfulness lemma (§5), and the substitutable, blocking, compound-path lookup (§6), which we have not found elsewhere. (Amin and Rompf's outcome space has no failure case; Jay and Given-Wilson's result concerns the limitations of SK over live terms rather than what an interpreter can choose to expose.)

3 The interpreter

3.1 Encoding and construction

The $\{S, K, I\}$ terms are the object language of the interpreter; in the terminology of Smullyan [1985] the standard definitions are:

$$\begin{aligned} \text{Idiot bird, } I &= \lambda x.x \\ \text{Kestrel, } K &= \lambda x.\lambda y.x \\ \text{Starling, } S &= \lambda x.\lambda y.\lambda z.x\,z\,(y\,z) \end{aligned}$$

Object terms are represented by Scott encoding [Scott, 1962, Jansen, 2013], in which a datum of an inductive type is a function that takes one continuation per constructor and applies the continuation for its own constructor to its own components. The type of object terms has four constructors, three leaves and application, so an encoded term takes four continuations, in the fixed order S, K, I, application:

$$\begin{aligned} \langle S\rangle &= \lambda s.\lambda k.\lambda i.\lambda z.\ s \\ \langle K\rangle &= \lambda s.\lambda k.\lambda i.\lambda z.\ k \\ \langle I\rangle &= \lambda s.\lambda k.\lambda i.\lambda z.\ i \\ \langle t\,u\rangle &= \lambda s.\lambda k.\lambda i.\lambda z.\ z\,\langle t\rangle\,\langle u\rangle \end{aligned}$$

A leaf discards all four continuations but its own. An application applies the fourth continuation to the encodings of its two children; this is how the interpreter walks a term, by applying $\langle t\rangle$ to four continuations as a case analysis on the head constructor of t. For example, $\langle K\rangle$ applied to four distinct atoms $X_1\,X_2\,X_3\,X_4$ reduces to X_2, and $\langle I\,K\rangle$ applied to the same four reduces to $X_4\,\langle I\rangle\,\langle K\rangle$. Bracket-abstracted into $\{S, K, I\}$, the encodings are closed combinators: $\langle K\rangle$ is the five atoms $K\,(S\,(K\,K)\,K)$; $\langle I\rangle$ is $K\,(K\,K)$; $\langle S\rangle$ is $S\,(K\,K)\,(S\,(K\,K)\,K)$; and $\langle I\,K\rangle$ is a 40-atom term which bears some discussion. The bracket abstraction of $\langle I\,K\rangle = \lambda s.\lambda k.\lambda i.\lambda z.\ z\,\langle I\rangle\,\langle K\rangle$ reads as three components composed together:

1. The first 32 atoms are the application constructor encA itself, $\lambda t.\lambda u.\lambda s.\lambda k.\lambda i.\lambda z.\ z\,t\,u$.
2. Three leading prefixes $S\,(K\,(S\,(K\,K)))$ push the three unused binders s, k, i past the body.
3. Finally, z is applied to t and u in the inner term, $\lambda skiz.k$.

The full expression thus reads,

$$\begin{aligned} &S \\ &(K(S(KK)))(S(K(S(KK)))(S(K(S(KK)))(S(S(KS)(S(KK)(S(KS)(S(K(SI))K))))(KK)))) \\ &(K(KK)) \\ &(K(S(KK)K)), \end{aligned}$$

which is depicted as a tree diagram in Figure 5, where it is discussed in context. When applied to any four terms, this expression hands $K(KK)$ and $K(S(KK)K)$, which are $\langle I\rangle$ and $\langle K\rangle$, to the fourth of them.

The encoding $\langle\cdot\rangle$ itself is a metalevel operation, for the reason given in §3.3. Lists, the `Maybe` of the result type, and the fuel numeral are Scott-encoded the same way:

- `nil` $= \lambda n.\lambda c.\ n$;
- `cons` $h\ t = \lambda n.\lambda c.\ c\,h\,t$;

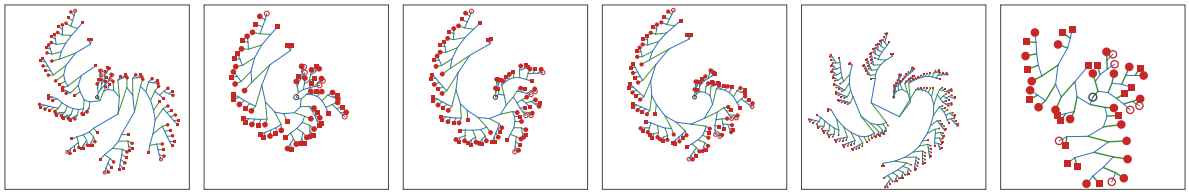


Figure 1: The interpreter's components as terms: the spine walker `sp` (126 atoms), the rebuilder `rb` (74), the three step arms `stepI` (92), `stepK` (105), `stepS` (237), and the unquoter `UQ` (43). Each panel is scaled to its frame, so sizes are not comparable across panels. Function edges are green and argument edges blue; S is a filled circle, K a square, I a ring. The S arm is substantially larger because it builds $\langle x\, z\, (y\, z)\rangle$ as data.

- `nothing` $= \lambda n.\lambda j.\, n$;
- `just` $x = \lambda n.\lambda j.\, j\, x$;
- `zero` $= \lambda z.\lambda s.\, z$; and
- `suc` $n = \lambda z.\lambda s.\, s\, n$.

Every definition below is written with named binders and expanded to a closed $\{S, K, I\}$ term by bracket abstraction, with recursion through the fixpoint combinator Y. The constructed interpreter, called `whnfF`, consists of a 618-atom term; its unquoter `UQ`, which maps encoded data back to a live term, contains 43 atoms. `whnfF` is given in full in Appendix B and depicted graphically in Figure 1. An evaluation is illustrated in Figure 2.

`whnfF` defines a fueled interpreter that evaluates encoded terms to weak head normal form. The fuel loop `whnfF n` $\langle t\rangle$ peels one `suc` from the numeral per step-attempt and returns `nothing` when the numeral is exhausted. Each attempt calls `step`, which walks the spine of the encoded term to its head leaf, collecting the arguments into a list, and dispatches on the leaf: the arm for I needs one argument, for K two, for S three. A saturated arm performs one contraction as data, building $\langle x\, z\, (y\, z)\rangle$ in the case of S, and the rebuilder `rb` re-applies the untouched arguments; the loop continues on the result with the remaining fuel. An unsaturated head means the term is in weak head normal form, and the loop returns it as `just` $\langle t\rangle$. This is normal-order head reduction with one contraction per attempt, which matches the host reducer's count exactly.

3.2 Self-application

The `whnfF` interpreter is applied to encoded terms to evaluate them to weak head normal form. This emulated process is not efficient: it incurs significant overhead due to the emulation of the host reduction steps. With fuel 3 on $\langle I\, K\rangle$ it reaches $\langle K\rangle$ in 340 host contractions; the decoded result equals the host's own reduction of $I\, K$. When it is run over the unquoter applied to an encoded term, it takes 91,556 steps. Run over an encoding of itself running on $\langle I\, K\rangle$, as a tower, it reaches weak head normal form in 504,930 contractions with the correct value. Empirically for this case, layer one costs 340 contractionsto emulate one, and layer two costs about 1,485 times layer one, the ordinary per-level overhead of naive self-interpretation. Figure 3 shows the first run in progress and Figure 4 the interpreter as a term.

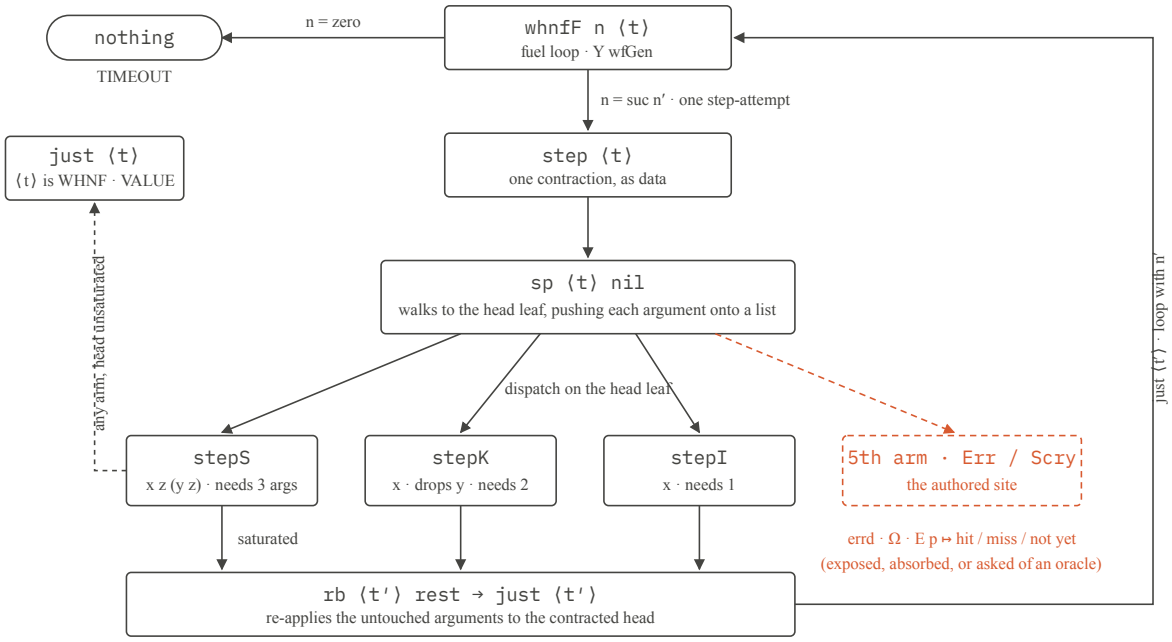


Figure 2: The `whnfF` evaluation process. The fuel loop peels one `suc` per step-attempt; `step` walks the spine to the head leaf, collecting the arguments; dispatch selects the arm for that leaf; a saturated arm contracts and `rb` re-applies the untouched arguments, returning `just` $\langle t' \rangle$ to the loop; an unsaturated head returns `just` $\langle t \rangle$, the value; exhausted fuel returns `nothing`. The dashed fifth arm optionally contains an authored site (§5).

3.3 The encoding boundary

$S\,K\,K$ and I agree on every argument, yet they are distinct normal forms as terms of weak combinatory logic. A quotation function from live terms to encoded data must be able to distinguish them, which cannot be done by any pure SKI term.

Proposition 3.1. *No closed term Q over $\{S, K, I\}$ satisfies $Q\,M \twoheadrightarrow \langle M \rangle$ for every closed M.*

Proof. Jay and Given-Wilson [2011, Thm. 3.2] prove that factorisation of SK-combinators is a symbolic computation not representable by any SK-combinator; the proof is stated for $\{S, K\}$ and goes through unchanged with I primitive. Quotation would give factorisation: Q together with the unquoter `UQ` of §3.1 decomposes any term into its head and arguments by Scott dispatch on $\langle M \rangle$. Directly: under the translation to the λ-calculus, $[\![Q]\!](\lambda x.x)$ is a single term, yet $Q\,(S\,K\,X)$ would yield $\langle S\,K\,X \rangle$, from which X is extractable, so every X would share a reduct, contradicting that S and K are distinct normal forms. □

Thus the encoder is a (better, the only) host function. Quotation is a one-time cost which must be paid to enter the representation. Everything on the other side of the boundary, the unquoter, the interpreter, their self-application, and the lookup machinery of §6 including equality on encoded paths, is ordinary $\{S, K, I\}$. Figure 5 shows the decoder beside a datum: in this calculus they are the same kind of thing.

4 The structural cost

What must any self-interpreter for SKI be able to do, structurally? S bundles two capabilities into one rule, composition and the duplication of its third argument, and never discards; K supplies discarding. The two capabilities that matter here, discarding and copying, coincide with the structural rules of weakening and contraction in substructural logic. By utilizing the basis $\{B, C, K, W\}$ for a comparative rewrite, we access the capabilities directly: B composes, C swaps, K discards an argument, and W duplicates an argument. We are interested in analyzing the structural cost of each component of the interpreter and the representation layer. Hence by rebuilding each component over the $\{B, C, K, W\}$ basis, we see which structural rule it requires, whether weakening (K) or contraction (W). We also count, independently of any abstraction algorithm, how often each binder occurs in each component's definition: a component needs weakening iff some binder occurs zero times, and contraction iff some binder occurs more than once. This occurrence census is the invariant measurement; atom counts are convention-relative (with I primitive the interpreter core shows seven W atoms, with $I = W\,K$ ten).

The representation layer is affine. All ten Scott constructors, the four term constructors together with `nil`, `cons`, `nothing`, `just`, `zero`, and `suc`, have no binder occurring twice: a constructor selects one continuation and discards the rest, which is weakening by construction. The representation lives in BCK.

The interpreter core needs contraction at five of eight components. There are seven surplus occurrences in all: the unquoter's f (twice, in $(f\,t)(f\,u)$), the spine generator's accumulator (four times), the rebuild generator's head (twice), the S-arm's z (twice, in $x\,z\,(y\,z)$, the S rule itself reappearing one level up), and the fuel loop's term (twice). Any $BCKW$ abstraction must perform at least that many contractions.

Each rule is needed, and one of them costs normalization. Removing weakening removes dispatch: a Scott constructor is an n-ary discard. Removing contraction removes iteration: Y duplicates its argument, and the affine fragment BCK is strongly normalizing, so it has no fixpoint combinator and no fuel loop. (The fuel itself, a Scott numeral, is affine; iteration comes from Y.) The two are not symmetric. Contraction alone already admits divergence $\perp$: $\Omega = (S\,I\,I)(S\,I\,I)$ contains no K, and in $BCKW$ the term $W\,W\,W$ reduces to itself with no

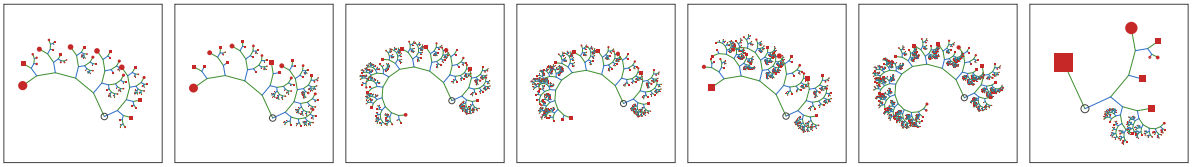
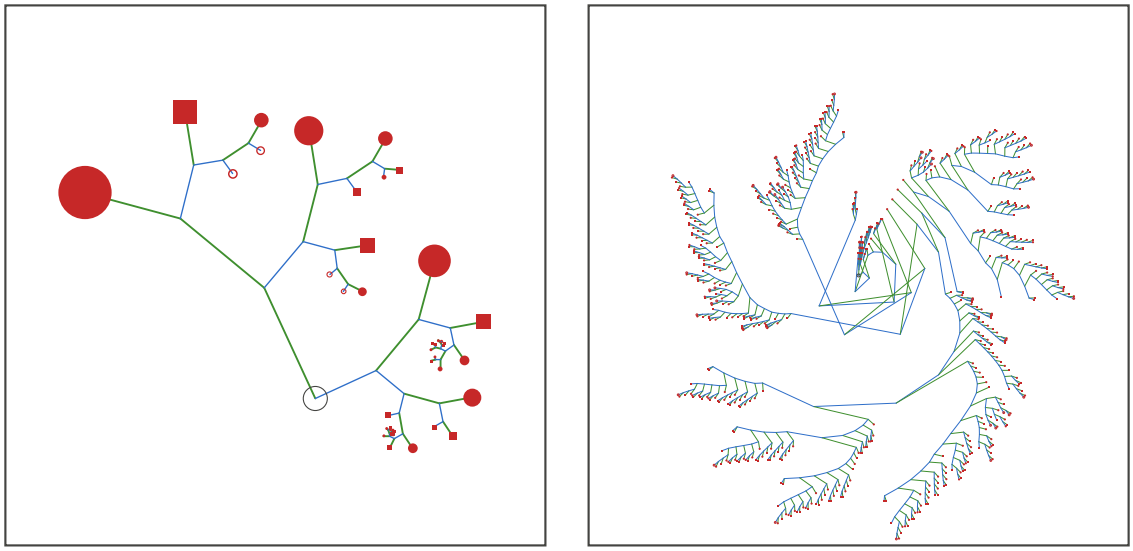


Figure 3: The interpreter running `whnfF 3` ⟨`I K`⟩, after 0, 20, 60, 120, 200, 300 and 340 host contractions, in the spiral view. The spine curls as Y unfolds and the encoded term is walked; the tree reaches about 111,000 atoms at step 300 and collapses to the answer, `just` ⟨`K`⟩, at step 340.



whnfF · spiral · 48 of 618 atoms

whnfF · every atom · 618

Figure 4: The interpreter `whnfF`: the angle is in-order leaf position, the radius is depth (which bottoms out at 48 levels). Each fern-like arm contains one deeply nested argument as produced by the bracket abstraction. Appendix B gives the term in full.

weakening anywhere, so the relevant fragment $BCIW$ is not normalizing while BCK is. Weakening is bought for representation and costs nothing in normalization; contraction is bought for iteration and is exactly what makes non-termination possible. Hence any self-interpreter for SKI must be written in a fragment where non-termination exists, and its outcome space cannot be {value}: it is at least {value, $\perp$}, which the next section makes observable by introducing fuel. The third outcome is not a design choice of this artifact but a consequence of self-interpretation.

The census also explains the artifact's shape. The representation is cheap because data only selects: every constructor and every case analysis is weakening, and the whole encoding of a term is built from discards. The cost is concentrated in five places, and all five are either recursion (the unquoter, the spine walker, the rebuilder, and the fuel loop each pass themselves or their state on twice) or the S rule reappearing one level up inside the arm that implements it, which is why that arm is the largest component (Figure 1). Two practical consequences follow: the representation layer can be carried unchanged into an affine calculus, with only the loop needing the full basis; and any effort to shrink the interpreter should be spent on those five sites, since everything else already bears the cost of a selection.

5 Extensional authored semantics

In a metacircular evaluator augmenting interpretation, additional outcomes can be observed and handled explicitly, rather than being hidden as divergence. This permits concepts like fuel and `Maybe` for partial evaluation to be introduced to a basis that without these concepts natively.

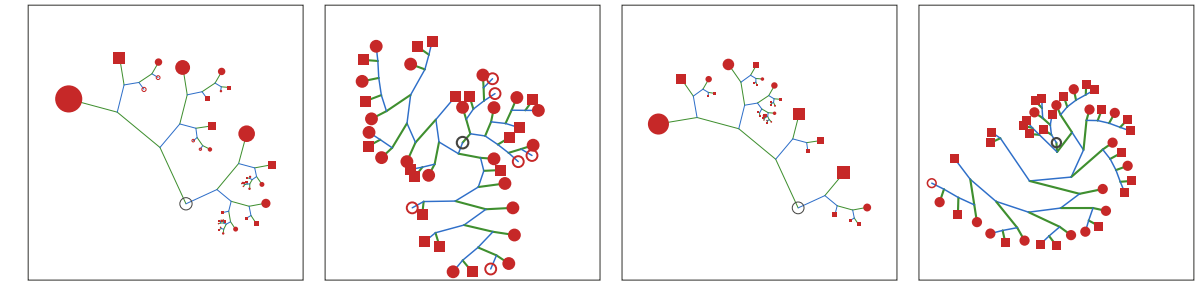


Figure 5: The decoder and a datum. `UQ`, 43 atoms, and the Scott encoding $\langle I\,K\rangle$ of the two-atom term $I\,K$, 40 atoms, each in spiral and every-atom views. A datum and a program look alike because they are the same kind of thing; only the encoding step that produced $\langle I\,K\rangle$ lies outside the calculus.

5.1 Fuel and the third outcome

An unfueled interpreter's outcome space is $\{\text{value}, \bot\}$, and divergence cannot be observed from inside a finite run. Fuel converts it to $\{\text{value}, \text{timeout}\}$, which are both observable. Such limitation is local and carries a clear price: at fixed fuel k, one observation does not separate a divergent term from one that needs more than k steps. Ω times out at every k; $I^{31}K$ times out for $k \leq 31$ and yields a value from $k = 32$, since fuel k permits k step-attempts of which the last must be the no-redex check. Convergence is semi-decidable by raising k while divergence is not (the halting problem). Object-level laziness is inherited, so $KI\Omega$ discards Ω unevaluated under the interpreter exactly as under host reduction because the object-level K rule fires before the interpreter recurses into the argument. (This follows Reynolds's observation.)

5.2 The site

Extend the object language with a fifth constructor `Err`, with no object-level contraction rule. Its encoding is unremarkable, five continuations instead of four. When the spine walker finds `Err` in head position, the interpreter decides how to dispatch:

```
D("errArmAbs", ["acc"], a("errd"))  # observable: crash becomes a value
D("errArmOmg", ["acc"], OMEGA)      # hidden: crash becomes host divergence
```

These arms are written $K\,(K\,(K\,I))$ and $K\,(S\,I\,I\,(S\,I\,I))$; our two full interpreters differ at this juncture. The result $\Omega = (S\,I\,I)(S\,I\,I)$ was selected as one of five divergent six-atom terms in this basis. The results of the two interpreters evaluating five different terms are summarized in Table 1.

Lemma 5.1 (Faithfulness)**.** *For every object term whose reduction never brings* `Err` *into head position, the exposing and hiding interpreters produce the same result.*

Proof. The two definitions differ only in the arm `errArm`, which is referenced only from the `Err` case of the step function, which the spine walker selects only when the head leaf is `Err`. On any other input the two are the same program. □

Table 1: Behavior of the two interpreters on selected object terms. The outcome "host diverges" refers to the step cap of the host reducer being reached, not a detected non-termination generally; the arm is Ω.

object term	exposing (`errArmAbs`)	hiding (`errArmOmg`)
`Err`	ERR	host diverges
`Err K`	ERR	host diverges
`K I Err`	VAL	VAL
`I K`	VAL	VAL
Ω	TIME	TIME

The `K I Err` row of Table 1 refers to an instance in which `Err` is present but discarded by object-level laziness before it is ever in head position. Here it highlights that the addition of a constructor does not by itself introduce a new meaning. The meaning is authored downstream by the author of the interpreter.

6 Site as a parameter

The substitution as written in §5.2 affords a choice between two source variants. A lookup in the style of Nock's "scry" or peek into a bound referentially transparent namespace takes the resolver as an argument instead. We build that in three steps, each extending the outcome type once more.

6.1 A synchronous resolver

The object language gains a fifth leaf `Scry`, of arity one, in a five-symbol encoding parallel to `Err`. The fueled loop gains a parameter: `wfQ E n` $\langle t \rangle$, where E is an ordinary encoded partial function from paths to values, a `Maybe`. When the spine walker reaches `Scry p` it applies E to p, the path exactly as written and unreduced. A hit splices the answer in and reduction continues. A miss is kept as a third step outcome distinct from "stuck", reusing the three-way type of §5.2, rather than folded into the existing `Maybe`, which would conflate "the resolver had no answer" with "no redex". The arm itself, in the same notation as the `Err` arms of §5.2, is

```
D("scHit",    ["rest","v"],     A(a("stepped"), A(a("rbQ"), v("v"), v("rest"))))
D("stepScQ1", ["e","p","rest"], A(v("e"), v("p"),
    a("errd"),                      # nothing    -> the miss
    A(a("scHit"), v("rest"))))      # just v     -> splice and continue
```

that is, apply the resolver to the path and hand it two continuations: on `nothing`, the miss; on `just v`, splice `v` in and rebuild. The parameter is the resolver E; when E is fixed at compile time to one answer, the two arms become the `Err` arms of §5.2.

The loop matches the base interpreter on every term without `Scry` against three resolvers (one that answers every path by echoing it, one that answers none, one that dispatches on the path's head leaf and answers only K); hits and misses where expected; reports a bare `Scry` as stuck rather than as either; reduces `K K (Scry K)` to K without consulting the resolver, so a site inside a branch the program never forces is never consulted; and times out on Ω before any site is reached. Substitutability is not eager; since the path is unreduced, `Scry (I K)` and

`Scry K` are different paths, a point to which §6.3 recurs. A path is not a sequence and not an address: it is whatever term occupies `Scry`'s argument position, delivered to the resolver as an encoded term, and it bears no meaning to the interpreter itself. Its interpretation is entirely the resolver's business. (Nock's scry paths are defined as lists of atoms; nothing here depends on that shape.)

6.2 A blocking resolver

A resolver that always answers is not a scry because a real lookup can fail to answer *yet* for a given path (such as a network call or result of a computation). The resolver's type grows from two-way to three-way (`oJust`, `oNothing`, `oNotYet`) and the interpreter's outcome types grow to match, adding `pending` at the step level and `block` at the result level. The arm gains one continuation:

```
D("stepScN1", ["e","p","rest"], A(v("e"), v("p"),
    A(a("scHitN"), v("rest")),      # oJust v    -> splice and continue
    a("errdN"),                     # oNothing   -> the miss
    A(a("pendingN"), v("p"))))      # oNotYet    -> block, carrying the path
```

The difference against the synchronous arm above is the third line. This extension triggers the blocking phenomenon of §5, providing one continuation at one site.

Resuming a block captures no continuation. Because reduction is pure and deterministic, the driver re-runs the whole term from the top under a resolver that now also answers the missing fact. Two conditions make that sound. The re-run's steps strictly before the block are identical to the first run's only if the new resolver agrees with the old one on every path the old one answered, which holds because the namespace is append-only. Re-running is not fuel-neutral: each round replays the previous one and then takes at least one more step, so with a fixed budget a block can become a timeout on the next round. Recomputation is charged for. The interpreter and each round's resolver are closed $\{S, K, I\}$ terms; the host side provides the fact table standing in for an external source of truth, the resume loop, and the detection of a path with no fact. (This starts to gesture towards a runtime handler which facilitates SKI evaluation in the context of blockable computations.)

The namespace is accumulative. With an empty namespace and facts $S \mapsto$ `Scry` K, $K \mapsto I$, the term `Scry S` blocks on S, learns it, re-runs, and the newly substituted `Scry K` blocks on K, a fact forced by the result of resolving the first, before the third round reaches the value I. A path with no fact blocks into a distinguishable stuck state, and fuel exhaustion remains distinguishable from a block throughout.

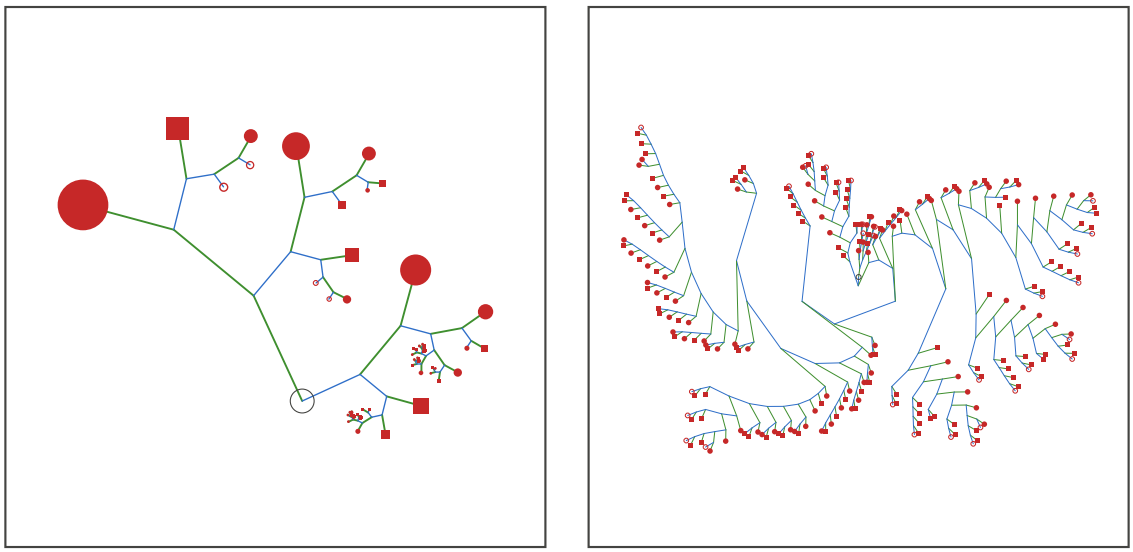
6.3 Compound paths

The resolver of §6.2 dispatches on a path's head leaf only and cannot tell $S\,K$ from $S\,K\,K$. Proposition 3.1 does not force this, as that theorem concerns factorising *live* combinators. The path a resolver receives is Scott-encoded *data*, a tree with five leaf tags, and equality on such data is an ordinary structural recursion. Jay and Given-Wilson said as much: once a term is encoded, factorisation "is a routine list operation". However, SK cannot produce the encoding from a live term.

We build `EQ5`, a structural equality test over the five-constructor encoding (Figure 6); and a resolver p that compares the incoming path against each known path in turn,

$$p \;\mapsto\; \texttt{EQ5}\ p\ \langle k_1 \rangle\ (\texttt{oJust}\ \langle a_1 \rangle)\ (\texttt{EQ5}\ p\ \langle k_2 \rangle\ (\texttt{oJust}\ \langle a_2 \rangle)\ (\cdots\ \texttt{oNotYet})),$$

which runs inside the blocking loop unchanged. This distinguishes a path from its prefix, resolving `Scry (S K K)` and `Scry (S K)` to different values under the same facts; it chains through a compound answer, `Scry (K I)` resolving to `Scry (S K)` and then to I in three rounds; and it refuses a false hit, blocking on `Scry (S K)` when the only fact is for $K\,S$.



EQ5 · spiral · 68 of 240 atoms

EQ5 · every atom · 240

Figure 6: `EQ5`, structural equality on encoded paths, 240 atoms: 92 S, 118 K, 30 I. Spiral view, 68 atoms visible, beside the every-atom disc.

Paths are intensional, syntactic objects, compared by structural equality of their encodings, exactly as a Nock scry path is a noun compared by noun equality. `Scry (I K)` blocks when the only fact is for K, and resolves when a fact is given for $I\,K$. Matching paths up to conversion constitutes the known-undecidable word problem for SKI; normalizing the path under fuel before lookup would be a fueled approximation thereto and would reintroduce the third outcome inside path resolution.

7 Discussion: virtualization and `+mock`

Nock's virtualized interpreter `+mock` is interesting not because Nock can interpret Nock; that follows once a formula computes `nock`, since a formula already is a noun. Its interest lies instead in that a formula can be run through a *substituted* interpreter, one that resolves a scry differently or catches a crash and returns it as a value, while agreement with the reference interpreter holds wherever the substitution is not exercised, and the substitution is confined to a named site. Read that way, §5 is the SKI analogue of Nock virtualization (rather than the tower of §3.2). The

tower demonstrates self-interpretation and fidelity, and a general version of that demonstration exists for value-producing towers [Amin and Rompf, 2018, Props. 4.2 and 4.6]. The two arms demonstrate the same thing `+mock` is for: the same program, the same interpreter apart from one arm, different observable outcomes in the case where the outcome categories differ. Lemma 5.1 asserts/requires faithfulness of the metacircular interpreter.

The analogy with Nock breaks at quotation. `+mock` never needs a quoting step, because a Nock formula already is the noun handed to the alternate interpreter. SKI is not homoiconic: S, K, I are opaque, and Proposition 3.1 states that no SKI term can turn an arbitrary SKI term into data about itself. SKI-in-SKI is therefore not symmetric the way Nock-on-Nock is. There is one unavoidable non-SKI quoting step, and virtualiation is native only above that boundary and closed under further quoting.

Depth is bounded for the ordinary reason. Two levels already cost 504,930 contractions against a trivial base term, at roughly 10^3 per level; a third level was not attempted in this research program. (Nock-on-Nock virtualization has the same character, which is why runtime-matched accelerated code exists for Nock as a practical instruction set architecture.)

8 Conclusion and open problems

An interpreter that is an ordinary term of the calculus it interprets makes the authorship of meaning visible: a new symbol's semantics is a few atoms at one site, provably isolated from everything else; that same site made into a parameter furnishes a namespace. Our construction is small enough to be read whole (Appendix B) and cheap enough to extend, and several open research questions follow therefrom:

- *Partial evaluation.* A resolver that treats an unanswered path as an inert unknown, rather than as a block, turns the blocking loop into a partial evaluator: run over an encoding of the interpreter applied to a fixed program with the resolver unknown, it returns the program with every dispatch before the first lookup executed away, Futamura's first projection in its weakest form. The static prefix is saved exactly; what remains after the first dynamic dependency is the interpreter's own continuation, and removing it requires unfolding under an unknown head, which is a separate project.
- *Quotation above level 0.* Proposition 3.1 bars quotation of live terms only. Re-encoding an already-encoded term, $\langle t \rangle \mapsto \langle\langle t \rangle\rangle$, is a data transformation and definable, and with it a tower can be deepened from inside; whether this recovers the self-quotation of Jay [2016, 2018] in the intensional setting is open.
- *An unquoter for the extended encoding,* and a pure-SKI driver for the blocking loop, would move the last host-side pieces of §6.2 into the calculus. The resolver's namespace is host-side only because its facts are compiled into a closure; an explicit core convention, a Scott pair of battery and payload in the manner of a Nock core, would make a fact an ordinary `cons` and the whole loop one closed term. That convention, and a subject-oriented surface language over it whose compile-time/runtime line for quotation is fixed by Proposition 3.1, are the subject of separate work. The semantics of intensional recursion [Kavvos, 2017] is the natural framework in which to state what such a driver would be.
- *Typed languages atop combinators.* The encoding of typed languages on top of combinatory logic, particularly in the presence of intensional features, can permit new kinds of type safety for second- and higher-order values. The Hoon language for the Nock combinator calculus exemplifies this but the space has not been extensively explored.

While SKI-on-SKI metacircular interpretation is hardly efficient, the development of a toolkit of reusable components and abstractions for SKI-on-SKI interpretation affords a promising structure for future exploration of combinator-based computing.

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A The artifact

All results were produced by the scripts in the accompanying repository against `aviary-kernel` 0.1.0, and every script checks its own claims: each printed result is compared against a fresh-marker probe of the reduced value rather than a printed outcome tag, and each script exits with a nonzero status if any check fails.

Files. `tower_harness.py` defines the interpreter and the two `Err` variants; `census.py` and `bckw_census.py` perform the census of §4; `scry_harness.py`, `scry_namespace.py`, and `scry_paths.py` build the three loops of §6, each loading the previous file's environment and modifying nothing in it; `figures.py` draws every figure from the expanded terms; `review/` holds the brute-force and construction checks cited in the text. `ski-in-ski.ipynb` executes end to end and reproduces the 618, 43, and 504,930 figures in its stored outputs; `ski-in-ski-aviary.ipynb` builds the interpreter natively in the `aviary` kernel as a walkthrough.

Sizes. Tree-atom counts, from `aviary_kernel.terms.size`:

term	atoms
`whnfF`, the interpreter	618
`UQ`, the unquoter	43
`wf5Abs` / `wf5Omg`, the two `Err` variants	768 / 771
`wfQ`, synchronous resolver loop	950
`wfN`, blocking resolver loop	1,066
`EQ5`, equality on encoded paths	240

The interpreter script's banner prints 348 and 62 for the first two; those are counts of DAG nodes, atoms plus application nodes deduplicated by identity, which is why the second exceeds 43.

Targets. `python3 tower_harness.py --target t0` (340 contractions), `t1` (91,556), `t2 --kk 500` (504,930; any object fuel of at least 341 gives this count, and the script refuses less), and `t3` (the table of §5.2). The three scry scripts take no arguments. The enumeration of divergent terms is `review/brute2.py`; the six-atom arm is `review/omg6.py`.

B The interpreter as an expression

The terms of Figures 4 and 5, written out. Application is juxtaposition and associates to the left; parentheses appear only where association is to the right; line breaks fall at spaces and carry no meaning. Both are exactly what bracket abstraction produces from the named definitions, so their atom counts are algorithm-relative in the sense of §4. Both begin with the same fourteen atoms, `S (K (S I I)) (S (S (K S) K) (K (S I I)))`, which is Y: each is a fixpoint of its generator.

`UQ`, the unquoter, 43 atoms:

```
S (K (S I I)) (S (S (K S) K) (K (S I I))) (S (K (S (S (S (S I (K S)) (K K))
(K I)))) (S (K K) (S (S (K S) (S (K (S (K S))) (S (K K)))) K)))
```

`whnfF`, the interpreter, 618 atoms:

```
S (K (S I I)) (S (S (K S) K) (K (S I I))) (S (K (S (S (K S) (S (K K) (S I (K
K)))))) (S (K K) (S (K (S (S (K S) (S (K K) (S (S (S (S (S (S (K (S I I)) (S
(S (K S) K) (K (S I I))) (S (K (S (S (K S) (S (S (K S) (S (S (K S) (S (S (K
S) K) (K (S (K (S (K K))) (S (K (S (K K))) (S (K (S I)) K)))))) (K (S (K K)
(S (K (S (K K))) (S (K (S I)) K)))))) (K (S (K K) (S (K K) (S (K (S I))
K)))))))) (S (K K) (S (S (K S) (S (K K) (S (K S) (S (K (S (K S))) (S (K
K)))))) (K (S (K K) (S (K (S (S (K (S (K K))) (S (S (K S) (S (K K) (S (K S)
(S (K (S I)) K)))) (K K)))))) K))))))) (K K)) (K (S (S I (K K)) (K (S (K (S
(S I (K K)))) (S (K K) (S (K (S (K (S (S I (K K)))))) (S (K (S (K K))) (S (K
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(S I I)) (S (S (K S) K) (K (S I I))) (S (K (S (S (K S) (S (K (S I)) K)))) (S
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(S (S (K S) (S (K K) (S (K S) (S (K (S I)) K)))) (K K)))))))) (K (S (K (S
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(K (S I)) K)))) (K K)))))))))))))))) (K (S (S I (K K)) (K (S (K (S (S I (K
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(K K))) (S (K (S (K K))) (S (S (K S) (S (K K) (S (K S) (S (K (S I)) K)))) (K
K))))))))))))))))) (K (S (S I (K K)) (K (S (K (S (K (S (K K) (S (K (S I))
K))))) (S (K (S I I)) (S (S (K S) K) (K (S I I))) (S (K (S (S (K S) (S (K (S
I)) K)))) (S (K (S (K K))) (S (K (S (K K))) (S (S (K S) (S (K K) (S (K S) K)))
(K (S (K (S (K K))) (S (K (S (K K))) (S (S (K S) (S (K K) (S (K S) (S (K
(S I)) K)))) (K K)))))))))))))) (S (K K) (S (K (S I)) K))))))) K)))
```